Available online at www.sciencedirect.com

SCIENCE DIRECT®

Cognitive Brain Research 22 (2005) 257–264

COGNITIVE BRAIN RESEARCH

www.elsevier.com/locate/cogbrainres

Research report

Intensive baseball practice improves the Go/Nogo reaction time, but not the simple reaction time

Noriyuki Kida[a], Shingo Oda[a], Michikazu Matsumura[b,*]

[a]*Laboratory of Human Motor Control, Graduate School of Human and Environmental Studies, Kyoto University, Sakyo-ku, Kyoto 606-8501, Japan*
[b]*Laboratory of Neuroscience, Graduate School of Human and Environmental Studies, Kyoto University, Sakyo-ku, Kyoto 606-8501, Japan*

Accepted 9 September 2004
Available online 18 October 2004

Abstract

Baseball hitters are required to make decisions whether to swing or not as quickly as possible. Therefore, we can assume that skilled baseball players have a quicker response. To verify this hypothesis, we assessed the effect of baseball experience or skill levels on simple reaction times and Go/Nogo reaction times in 82 university students (22 baseball players, 22 tennis players, and 38 nonathletes) and 17 professional baseball players. Also, to clarify whether this ability was innate or acquired, we examined the effects of long-term practice for baseball hitting in 94 senior high school students (26 baseball players and 68 non-baseball players). There were no differences in simple reaction time either for sports experience or for skill levels. On the contrary, the Go/Nogo reaction time for baseball players was significantly shorter than that of the tennis players and nonathletes. The Go/Nogo reaction time of higher-skill baseball players was significantly shorter than that of lower-skill players, while that of the professional baseball players was the shortest. The professional players showed the highest (almost linear) correlation between the Go/Nogo reaction time and simple reaction time. A longitudinal study showed that 2 years of hitting practice improved the Go/Nogo reaction time, while the simple reaction time remained constant. A cross-sectional study of high school non-baseball players showed no differences either in the simple or Go/Nogo reaction times in relation to school year. In conclusion, intensive practice, including Go or Nogo decision making, improved the Go/Nogo reaction time, but not the simple reaction time.
© 2004 Elsevier B.V. All rights reserved.

Theme: Neural basis of behavior
Topic: Cognition

Keywords: Reaction time; Go/Nogo task; Baseball; Skill level; Development of reaction time; Motor learning

1. Introduction

It is thought that reaction time is a sensitive measure of reaction, with simple reaction time being a measure of the overall speed of the perceptual and motor systems [11,20]. Because athletes are required to make quick decisions in daily training, long-term practice in sports may develop the speed of the perceptual and motor systems, and hence a simple reaction time may be able to predict the processing speed of the perceptual and motor systems of athletes. In previous experimental studies, however, it was pointed out that there was no relationship between simple reaction time and offensive ability in baseball [16] or other sports [1,12]. In other words, long-term practice in sports did not improve simple reaction abilities.

Athletes, especially those with open skills (see Schmidt and Lee [20]), have many decisions to make as they compete. We can assume that sports practice improves some reaction time abilities, especially sports-specific decision-making abilities. For example, baseball hitters are required to make decisions whether to swing or not as quickly as possible. This type of performance is known as a Go/Nogo reaction task, a choice of reactions, which has been frequently used to assess volitional inhibition in neurophysiological studies [8–10,13,14]. In a Go/Nogo

* Corresponding author. Tel.: +81 75 753 6889; fax: +81 75 753 6876.
E-mail address: matsumura@neuro.mbox.media.kyoto-u.ac.jp (M. Matsumura).

0926-6410/$ - see front matter © 2004 Elsevier B.V. All rights reserved.
doi:10.1016/j.cogbrainres.2004.09.003

task, subjects are required to make a specific motor response to one class of stimuli (Go response), and to withhold the Go response to the other (Nogo response). The voluntary decision ability to do or inhibit an already prepared action is required for baseball hitters, who have the advantage of withholding a swing at the ball in the 'ball zone'. We can assume that long-term practice for baseball hitting may develop Go or Nogo decision-making abilities. In other words, the Go/Nogo reaction time may be able to correctly predict the ability of quickness of judgment necessary for actual baseball hitting. Although the simple and choice reaction times of baseball players were investigated in previous studies [5,16], there are few studies dealing with the Go/Nogo reaction time for baseball players.

We hypothesized that (1) baseball hitters are so much more accustomed to Go/Nogo reaction tasks that they can quickly make Go or Nogo decisions. To verify this hypothesis, we assessed the effects of baseball experience on Go/Nogo and simple reaction time. We also assume that (2) more highly skilled baseball players may have shorter Go/Nogo reaction times compared with lower skilled baseball players. To verify this hypothesis, we assessed the effects of skill levels on Go/Nogo and simple reaction times. We also focused on the impact of practice on the performances of identical baseball players of high school age in a longitudinal study to clarify whether higher skilled baseball players are innately better at performance of Go/Nogo tasks.

2. Materials and methods

2.1. Subjects

Seventeen male professional baseball players (AAA level, 23.4±2.1 years old), 82 male university students (22.1±1.9 years), and 94 male senior high school students volunteered to take part in this study. The university students made up three subgroups; 22 baseball players (21.6±2.0 years) who belonged to the official University team in the Japanese Kansai Big 6 Baseball League, 22 tennis players (21.3±1.4 years) who belonged to the official university team in the Japanese Kansai Student Tennis League, and 38 sedentary students (22.9±1.7 years) who had no special experience of any kind of ball game training. The university baseball players and tennis players were further divided into three sub-groups according to their skill levels. High-skill players were regular members, medium-skill players were substitutes, and low-skill players were members in name only (seven, eight and seven players, respectively).

The high school students made up two subgroups; 26 baseball players who belonged to the official high school team in the Japan High School Baseball Federation and 68 non-baseball players who had no experience of intensive hitting baseball practice. The high school baseball players participated in the same experiment three times in the first, second, and third school years (15.3±0.5, 16.3±0.5, and 17.3±0.5 years). The non-baseball players were divided into three subgroups according to their school year. Thirty-one students were in the first year (15.2±0.4 years), 23 were in the second year (16.3±0.5 years), and 14 were in the third year (17.3±0.3 years). To examine the learning effect by repeating the same experiment, 14 of the 68 non-baseball players participated in the same experiment three times at five months intervals. The subjects were all right handed. Informed consent was obtained from each subject according to the approval of the Ethical Committee of the Graduate School of Human and Environmental Studies, Kyoto University.

2.2. Procedure

The subjects sat on a comfortable chair in a dimly illuminated room. Visual stimuli were presented on a computer screen located about 50 cm in front of the subject. Four square frames (3×3 cm) were presented in a horizontal row on the screen. Their horizontal visual range was 27.4 degrees. The frames were sequentially called 'frame 1', 'frame 2', 'frame 3', and 'frame 4' from the left. Two different behavioral tasks were addressed in the present study: a simple reaction time task and a Go/Nogo reaction time task. In the simple reaction time task, one of the four square frames was colored green at random. The subjects responded to the color square presentation as rapidly as possible, irrespective of the stimulus location by pressing a keyboard with their right index finger. In the Go/Nogo reaction time task, the subject was asked to respond whenever either of the two inside frames (Stimuli 2 and 3: corresponding to the 'strike zone') colored green (Go trial), but to withhold the response to the outside frames (Stimuli 1 and 4: corresponding to the 'ball zone').

Each task consisted of three blocks; the first block contained 20 trials and the second and third blocks 40 trials, respectively. Thus, the total number of trials in each task was 100. Subjects took short rests between the blocks and a longer rest between the simple and Go/Nogo tasks. Before each reaction task, 20 practice trials were performed to familiarize the subject with the new task. These tasks were performed at random and were counter-balanced. The duration of all visual stimuli was 50 ms and inter-trial intervals were either 3, 4, or 5 s. The orders of stimulus locations and inter-trial interval were randomized. In any task conditions, therefore, the probability of stimulus appearance at each location was 25%.

2.3. Behavioral measurements and analyses

All procedures for stimulus presentation and measurements of reaction times were controlled by a desktop

computer (PC-9821; NEC, Tokyo, Japan). Reaction time was determined as the time from the stimulus onset to the key-press response. Trials with reaction time shorter than 100 ms were regarded as anticipation errors and excluded from the analyses (0.5% of all trials). Responses to the simple and Go stimuli with reaction times of more than 1000 ms were regarded as omission errors and were excluded from the analyses (1% of all trials). In the Go/Nogo task, when the subject responded to Stimulus 1 or 4 (Nogo signal), the trial was evaluated as a commission error.

3. Results

3.1. Simple reaction time

Fig. 1 represents group data for simple reaction time showing a comparison for sports experience and skill level. One-way analysis of variance showed that there were no significant differences in the mean simple reaction time for university students among three different groups of sports experience (for nonathletes, tennis, and baseball players, 245±24, 243±23, and 241±24 ms, means±S.D.s, respectively, $F(2,79)=0.25$, ns). In order to examine the impact of skill levels on simple reaction times, the baseball and tennis players were further divided into three subgroups according to their skill levels. One-way analysis of variance with skill level showed that there were no significant differences in simple reaction time in either group (baseball players, $F(2,19)=1.14$, ns; tennis players, $F(2,19)=0.17$, ns). Including the professional baseball players, the difference among the four baseball subgroups was not significant ($F(3,35)=0.93$, ns).

3.2. Go/Nogo reaction time

Fig. 2 shows the reaction time in the Go trials of the Go/Nogo task. One-way analysis of variance showed that there was a significant difference in the Go reaction time for university students among the three groups of sports experience ($F(2,79)=10.01$, $p<0.01$). Multiple comparisons, using the Scheffe procedure, indicated that reaction times of the baseball players (293±37 ms) were significantly shorter than those of the tennis players (332±34 ms, $p<0.05$) and nonathletes (347±46 ms, $p<0.01$).

To examine the impact of skill level on the Go/Nogo reaction times, the baseball and tennis players were similarly subdivided into three subgroups. As for baseball players, there were significant differences among three different groups in relation to skill levels ($F(2,19)=4.00$, $p<0.05$). The Go/Nogo reaction time of high-skill players (276±38 ms) was significantly shorter than that of the low-skill players (323±35 ms, $p<0.05$). Including professional baseball players, the differences among the four subgroups became more prominent ($F(3,35)=6.68$, $p<0.01$). The Go/Nogo reaction time of the professional players (263±31 ms) was significantly shorter than that of the medium- (293±37 ms, $p<0.05$) and low-skill players ($p<0.01$). For tennis players, however, there were no differences in the Go/Nogo reaction time among the three different skill levels ($F(2,19)=0.93$, ns).

3.3. Error rates

To clarify whether the facilitation of the Go/Nogo reaction time was the result of a speed-accuracy trade-off, commission errors were analyzed in these groups. Fig. 3 represents the group data of the commission error rate for

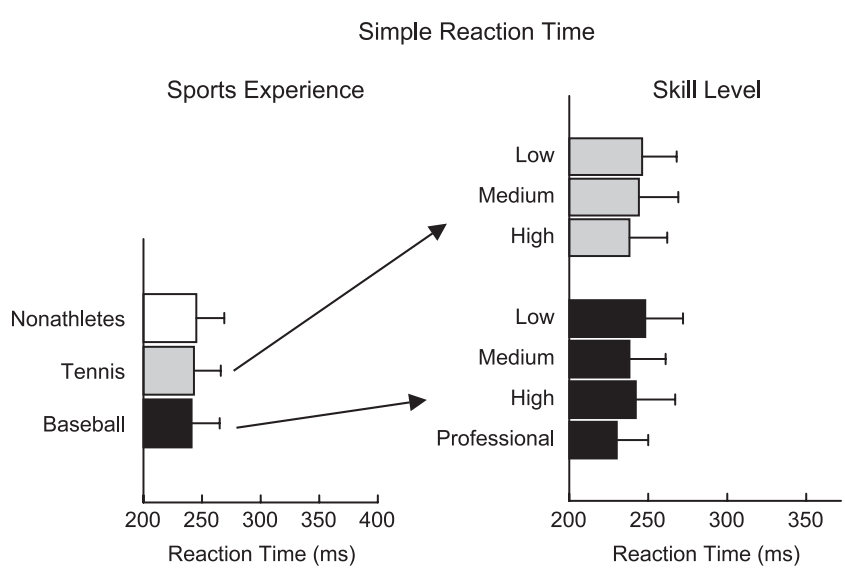

Fig. 1. Simple reaction time (mean±S.D.) did not represent performance abilities. There were no differences in reaction time among students in the three different groups of sports experience; nonathletes, tennis and baseball players (left). There were also no differences in simple reaction time among different groups in relation to skill levels among baseball and tennis players (right).

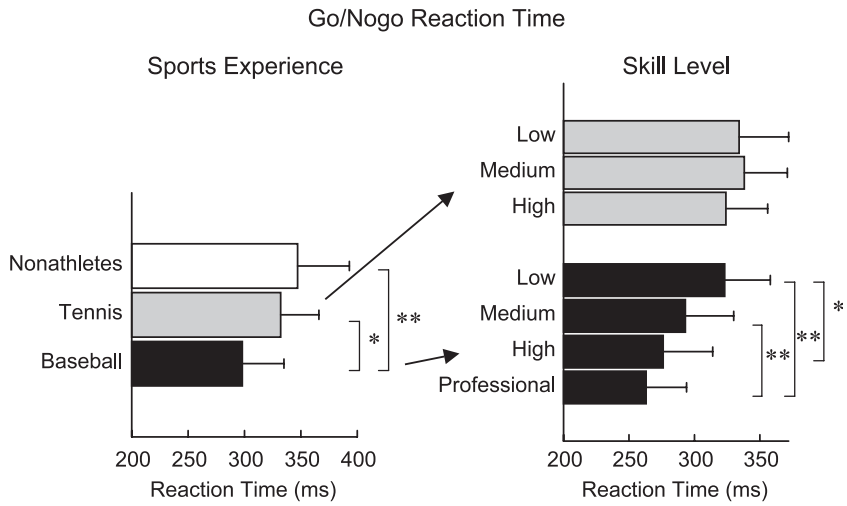

Fig. 2. Shorter Go reaction time (mean±S.D.) for baseball players and the shortest Go reaction time in high-skill level baseball players. The Go/Nogo reaction time for baseball players was significantly shorter than that for the tennis players and nonathletes (left). The Go/Nogo reaction time of higher-skill baseball players was significantly shorter than that of lower-skill players, whereas there was no difference among the three groups of skill levels for tennis players. The professional baseball players had the shortest Go/Nogo reaction time (right). *$p<0.05$, **$p<0.01$.

sports experience and skill level. There were no differences in mean commission error rate for university students among the three different groups of sports experience ($F(2,79)=0.93$, ns). In both tennis and baseball players, there were no differences in commission error rate among the three different skill levels either (baseball players, $F(2,19)=0.73$, ns; tennis players, $F(2,19)=0.38$, ns). Including the professional baseball players, the difference among the four subgroups was not significant ($F(3,35)=0.98$, ns).

3.4. Age differences (longitudinal and cross-sectional study) of Go/Nogo reaction time

Fig. 4 shows the simple reaction time and Go/Nogo reaction time for the high school students in the longitudinal and cross-sectional studies. The longitudinal study for high school baseball players showed that there was no difference in the simple reaction time ($F(2,50)=2.32$, ns). However, there was a significant difference in the Go/Nogo reaction time ($F(2,50)=10.57$, $p<0.01$). Multiple comparisons, using the Scheffe procedure, showed that the Go/Nogo reaction time in the third-year (289±24 ms) decreased significantly compared to the first-year (300±26 ms, $p<0.01$) and second-year (296±27 ms, $p<0.01$). On the contrary, the cross-sectional study for high school non-baseball players showed that there were no differences in either the simple reaction time ($F(2, 65)=1.40$, ns) or the Go/Nogo reaction time ($F(2, 65)=0.41$, ns).

One-way analysis of variance of subjects' sports experience was performed for each school year (Fig. 4). The

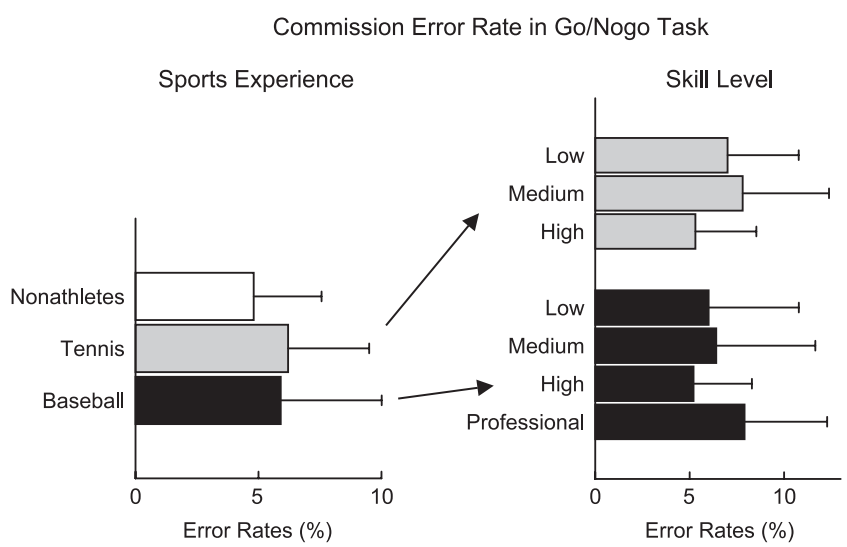

Fig. 3. Commission error rate (mean±S.D.) in the Go/Nogo reaction time task. There were no differences in commission error rates for student among the three different groups of sports experience (left). There were also no differences in error rates among different skill levels in baseball and tennis players (right).

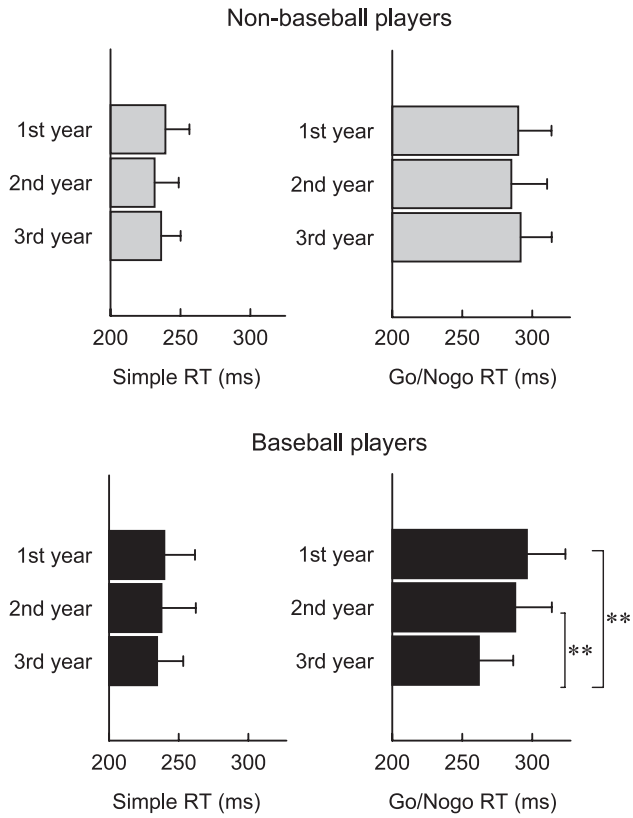

Fig. 4. The simple reaction time and Go/Nogo reaction time for the high school students. For baseball players, although there was no difference in the simple reaction time (top-left), the Go/Nogo reaction time in the third year was shorter than that in the first and second years (top-right). On the contrary, for non-baseball players, there were no differences in the simple reaction time (bottom-left) and Go/Nogo reaction time (bottom-right). $^{**}p<0.01$.

results showed that there were no differences in the Go/Nogo reaction time between baseball and non-baseball players in the first year ($F(1,55)=0.840$, ns) and the second year ($F(1,47)=0.137$, ns). In the third year, however, the Go/Nogo reaction time of baseball players was significantly shorter than that of non-baseball players ($F(1,25)=14.259$, $p<0.01$).

We examined the test–retest improvement three times in the non-baseball high school students using the simple and Go/Nogo reaction time task. The results showed that there were no significant differences among the three repeated experiments in either the simple reaction time (first, second, and third time, 239±22, 243±15, and 238±19 ms, means±S.D.s, respectively, $F(2,13)=0.81$, ns) or the Go/Nogo reaction time (first, second, and third time, 328±23, 337±28, and 333±30 ms, respectively, $F(2,13)=0.77$, ns).

3.5. Correlation analysis between the simple and Go/Nogo reaction times

As shown in Fig. 5, the Go/Nogo reaction time was significantly correlated with the simple reaction time in all groups ($p<0.01$). The university baseball players ($r=0.85$) had a higher correlation than nonathletes ($r=0.61$) and tennis players ($r=0.69$). The professional baseball players ($r=0.99$) showed the highest (almost linear) correlation. For high school baseball players, the third-year students ($r=0.83$) had a higher correlation between the simple and Go/Nogo reaction times than first-year students ($r=0.62$).

4. Discussion

4.1. Previous simple reaction time hypothesis

Research models that view athletes as processors of information have become very popular [11,20]. Researchers have proposed various models and believe that many different information-processing activities occur during the time between the presentation of a stimulus and completion of a response. It is thought that reaction time is a sensitive measure of reaction, with simple reaction time being a measure of the overall speed of the perceptual and motor systems. Therefore, it was natural to assume that simple reaction time might be able to predict the processing speed of the perceptual and motor systems of athletes. In previous experimental studies, however, Nielsen and McGown [16] demonstrated no relationship between simple reaction time and offensive ability in baseball. Mero [12] reported no significant difference in simple reaction time between novice and expert tennis players. Not only with baseball or tennis players, but also with clay shooters [1], basic visual functions, including simple reaction time, were not a limiting factor to sports performance. In the present study, we confirmed that there was no difference in the simple reaction time among the three different groups of sports experience and among the three groups of skill levels for both baseball and tennis players. According to the present study and the above previous studies [1,12,16], we conclude that simple reaction time is not a particularly accurate predictor of experience, performance, or success in sports.

4.2. Shorter Go reaction time for baseball players

In the present study, the Go reaction time of the baseball players was significantly shorter than that of the tennis players and nonathletes. For baseball players, the Go reaction time of higher-skill players was significantly shorter than that of lower-skill players. For tennis players, however, there were no differences in the Go/Nogo reaction time among the three different skill levels. Previous studies [4,6,21,23] reported that subjects who were more deliberate in identifying the appropriate response showed a longer reaction time in Go/Nogo tasks with a loss of speed in decision making. This leads to speculation that the longer reaction time in the present Go/Nogo task resulted from greater deliberation in identifying the appropriate response. Therefore, if baseball players in the present study had tried to make quicker decisions by sacrificing accuracy, the

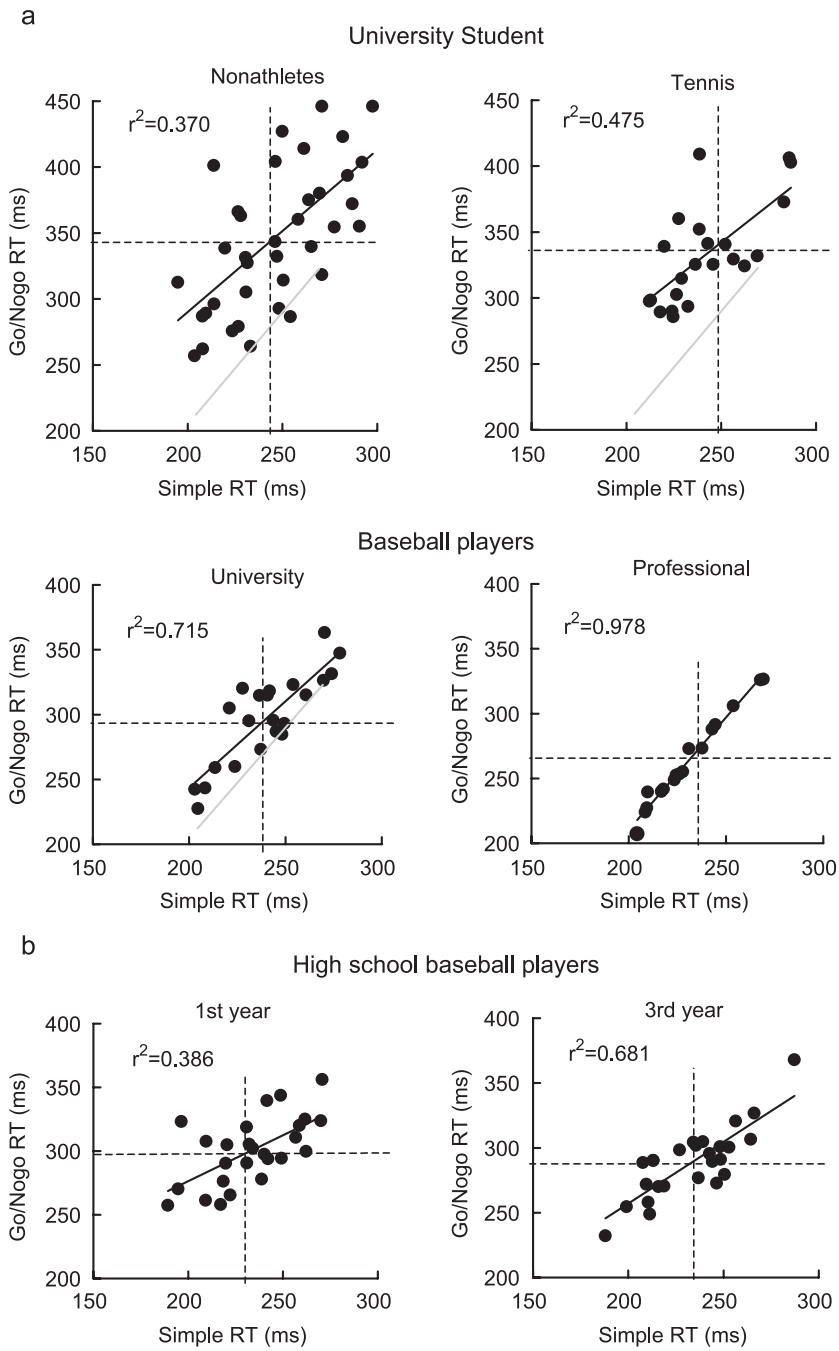

Fig. 5. High correlations between the simple reaction time and Go/Nogo reaction time in the skillful baseball players. (a) Relationship between the simple reaction time and Go/Nogo reaction time of university students and professional baseball players (about the same age). Dashed lines indicate the mean reaction times. The nonathletes (top-left) and tennis players (top-right) show a weaker correlation than the baseball players (middle-left). The professional baseball players (middle-right) show the highest correlation. Gray lines were identical regression lines for the professional players. (b) Longitudinal measurements of reaction times for high school baseball players in the first-year (bottom-left) and third-year (bottom-right). The correlation became stronger after 2 years of hitting practice.

decision errors of the baseball players would have been higher than those of the nonathletes and tennis players. In fact, there was no significant difference in commission error rates among the three different groups in relation to sports experience. This indicates that the difference in Go/Nogo reaction time did not result from the difference in deliberation in identifying the appropriate response. Furthermore, there were no differences in the commission error rates among the baseball players with different skill levels. This indicates that the quicker Go/Nogo reaction performances for highly skilled baseball players did not sacrifice accuracy in Go or Nogo decision making. Taken together, we conclude that a 'speed-accuracy trade-off [7]' was not observed in this experiment.

It is not clear whether highly skilled baseball players are innately better at performance of Go/Nogo tasks. Therefore, we focused on the impact of practice on performances of identical baseball players of high school age. According to the previous studies for development and aging, simple reaction times gradually decrease by the middle-teens, and hold constant at a stable value until middle age [2,24,25]. Our longitudinal study of high school baseball players showed that an intensive 2 years of baseball practice improved the Go/Nogo reaction time, while the simple reaction time again remained constant. On the contrary, the cross-sectional study for the high school non-baseball players showed that there were no differences in the simple and Go/Nogo reaction times among the three groups in relation to their school year. For the high school non-baseball players, there were no differences among three repeated experiments either in the simple and Go/Nogo reaction times. Thus, it can be assumed that the decreased Go/Nogo reaction time in the high school baseball players in the third-year did not include learning effects by repeating the Go/Nogo reaction time experiment in the present study.

There was no difference in the Go/Nogo reaction time between the baseball players and non-baseball players either in the first or second year. These findings suggest that the shorter time needed to make Go or Nogo decisions for the baseball players is not innate. In other words, the neural processing required for Go or Nogo decision making can improve with baseball practice. Many researchers have reported that both simple and choice reaction times decrease with practice [3,17,18,22]. According to these studies, it is likely that reaction time decreases with laboratory practice. However, there is no study investigating whether baseball experience has an effect on laboratory reaction time task. The present study showed that the Go/Nogo reaction time recorded in the laboratory decreased through baseball experience. This finding suggests that skills acquired by playing baseball can be transferred to a laboratory Go/Nogo reaction time task. However, it remains unknown whether these baseball players, who were already good enough at baseball to make the team, had a greater capacity to improve the Go/Nogo reaction time than non-baseball players.

According to event-related potential studies, a P300 component is related to Nogo trials, especially the response inhibition process [6,19]. Jodo and Inoue [8] reported that the Go/Nogo reaction time and P300 latency to Nogo stimuli were significantly shortened by intensive practice for 6 days, suggesting learning effects. From these findings and the present study, it can be speculated that baseball players have developed higher abilities to quickly perform the Go reaction via long-term baseball training. We also found that the shorter Go/Nogo reaction time for the high-skill baseball players can be attributed to long-term training, leading to the increased motor skills required for high-level baseball performance. From this perspective, we conclude that the professional baseball players in the present study had improved their Go or Nogo decision making, and thus had the shortest Go/Nogo reaction time. The Go/Nogo reaction time could be a skill level index for baseball players.

4.3. Proposal of a Go/Nogo learning hypothesis for baseball players

This longitudinal study of high school baseball players showed that the correlation between the simple and Go/Nogo reaction time was strengthened by 2 years of baseball practice, indicating that the 2 years of baseball practice decreased the variance of the central processing time needed for Go or Nogo decision. These findings suggest that the Go or Nogo decision time could be shortened toward a possible minimum value of each individual player. For university students and professional baseball players as well as high school students, the longer the baseball experience, the stronger the correlation between the simple and the Go/Nogo reaction times. Moreover, the professional baseball players showed an almost linear correlation between them. It appears that the Go/Nogo reaction times are a function of the simple reaction times. From the results of the professionals, the mean Go reaction time (Tg) for the most trained players can be predicted by the following equation

$$Tg \text{ (in milliseconds)} = 1.7\text{Ts} - 135$$

where Ts represents mean simple reaction time of each subject. This indicates that the Go reaction time for the best-trained subjects is determined only by a factor of the simple reaction time. In other words, simple reaction time limits the Go/Nogo reaction time. These results are considered to have important consequences both from practical and theoretical stand points. On the practical side, long-term intensive training for Go or Nogo decision can shorten the Go reaction times to the point where the simple reaction times of individual players have been set. It is suggested that simple reaction times can be used as predictors showing to what extent the Go/Nogo reaction time of baseball players can be shortened through intensive baseball practice.

On the theoretical side, the mechanism underlying our results that the simple reaction time limits the Go/Nogo reaction time may contribute to further brain studies. According to recent positron emission tomography studies or functional magnetic resonance imaging studies, Naito et al. [15] showed that the anterior cingulate cortex was a key structure which determined the speed of response in reaction time tasks. Liddle et al. [10] showed that during Go/Nogo reaction tasks, the anterior cingulate cortex is principally engaged in making and monitoring Go or Nogo decisions, while the dorsolateral and ventral lateral prefrontal sites play a specific role in response inhibition. Based on the these brain studies, our behavioral study may show that a task which is believed to initially be under the control of above central nervous processes can be turned into a simple reaction time plus a constant.

Hence, the following conclusions are derived: (1) Simple reaction time is not a particularly accurate predictor of

experience, performance, or success in sports. (2) Go or Nogo decision time can be improved by baseball practice, and is not innately fixed. In contrast, there was no impact of baseball practice on simple reaction times in adults. (3) The Go/Nogo reaction time could be theoretically shortened toward a certain value determined by the simple reaction time proper to each individual.

References

[1] B. Abernethy, R.J. Neal, Visual characteristics of clay target shooters, J. Sci. Med. Sport 2 (1999) 1–19.
[2] K. Andersen, L. Starck, I. Rosen, E. Svensson, The development of simple acoustic reaction time in normal children, Dev. Med. Child Neurol. 26 (1984) 490–494.
[3] S. Ando, N. Kida, S. Oda, Practice effects on reaction time for peripheral and central visual fields, Percept. Mot. Skills 95 (2002) 747–751.
[4] J. Botwinick, L. Thompson, Premotor and motor components of reaction time, J. Exp. Psychol. 71 (1966) 9–15.
[5] J.G. Classe, L.P. Semes, K.M. Daum, R. Nowakowski, L.J. Alexander, J. Wisniewski, J.A. Beisel, K. Mann, R. Rutstein, M. Smith, A. Bartolucci, Association between visual reaction time and batting, fielding, and earned run averages among players of the southern baseball league, J. Am. Optom. Assoc. 68 (1997) 43–49.
[6] M. Eimer, Effects of attention and stimulus probability on ERPs in a Go/Nogo task, Biol. Psychol. 35 (1993) 123–138.
[7] P.M. Fitts, The information capacity of the human motor system in controlling the amplitude of movement, J. Exp. Psychol. 47 (1954) 381–391.
[8] E. Jodo, K. Inoue, Effects of practice on the P300 in a Go/NoGo task, Electroencephalogr. Clin. Neurophysiol. 76 (1990) 249–257.
[9] R. Kawashima, K. Satoh, H. Itoh, S. Ono, S. Furumoto, R. Gotoh, M. Koyama, S. Yoshioka, T. Takahashi, K. Takahashi, T. Yanagisawa, H. Fukuda, Functional anatomy of GO/NO-GO discrimination and response selection—a PET study in man, Brain Res. 728 (1996) 79–89.
[10] P.F. Liddle, K.A. Kiehl, A.M. Smith, Event-related fMRI study of response inhibition, Hum. Brain Mapp. 12 (2001) 100–109.
[11] Magill, Motor Learning: Concepts and Applications, Brown and Benchmark, Dubuque, 1999.
[12] A. Mero, L. Jaakkola, P.V. Komi, Neuromuscular, metabolic and hormonal profiles of young tennis players and untrained boys, J. Sport Sci. 7 (1989) 95–100.
[13] J.O. Miller, K. Low, Motor processes in simple, Go/No-go, and choice reaction time tasks: a psychophysiological analysis, J. Exp. Psychol. Hum. Percept. Perform. 27 (2001) 266–289.
[14] E. Naito, M. Matsumura, Movement-related potentials associated with motor inhibition under different preparatory states during performance of two visual stop signal paradigms in humans, Neuropsychologia 34 (1996) 565–573.
[15] E. Naito, S. Kimura, S. Geyer, R. Kawashima, P.E. Roland, K. Zilles, Fast reaction to different sensory modalities activates common fields in the motor areas, but the anterior cingulate cortex is involved in the speed of reaction, J. Neurophysiol. 83 (2000) 1701–1709.
[16] D. Nielsen, C. Mcgown, Information processing as a predictor of offensive ability in baseball, Percept. Mot. Skills 60 (1985) 775–781.
[17] R.W. Proctor, T.G. Reeve, The acquisition of task-specific productions and modifications of declarative representations in spatial-precuing task, J. Exp. Psychol. Gen. 117 (1988) 182–196.
[18] R.W. Proctor, T.G. Reeve, D.J. Weeks, L. Dornier, T. Van Zandt, Acquisition, retention, and transfer of response selection skill in choice reaction tasks, J. Exp. Psychol. Learn. 17 (1991) 497–506.
[19] L.E. Roberts, H. Rau, W. Lutzenberger, N. Birbaumer, Mapping P300 waves onto inhibition: Go/No-Go discrimination, Electroencephalogr. Clin. Neurophysiol., Evoked Potentials 92 (1994) 44–55.
[20] R.A. Schmidt, T.D. Lee, Motor Control and Learning, Human Kinetics, Champaign, 1999.
[21] I. Silverman, Age and the tendency to withhold a response, J. Gerontol. 18 (1963) 372–375.
[22] Y. Taniguchi, Effect of practice in bilateral and unilateral reaction-time tasks, Percept. Mot. Skills 88 (1999) 99–109.
[23] A.T. Welford, Motor performance, in: J.E. Birren, K.W. Schaie (Eds.), Handbook of the Psychology of Aging, Van Nostrand Reinhold, New York, 1977.
[24] B.R. Williams, J.S. Pnesse, G.D. Logan, R. Tannock, Development of inhibitory control across the life span, Dev. Psychol. 35 (1999) 205–213.
[25] A.V. Zaitsev, V.I. Lupandin, O.E. Surnina, Age-related dynamics of the time of reaction to visual stimuli, Hum. Physiol. 25 (1999) 656–659.